Xuhao Luo

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Education

University of Illinois Urbana-Champaign Aug. 2021 - Nov. 2026 (Anticipated)
Ph.D. Candidate in Computer Science
University of California San Diego Sep. 2019 - Mar. 2021
M.S. in Computer Science
University of Science and Technology of China Sep. 2015 - Jun. 2019
B.S. in Applied Physics

Interests

Storage, Memory, and Network Systems for AI (High-performance storage, Multi-tier KV-Cache, GPUDirect storage, etc.)

Research Publication

- **Xuhao Luo**, Emaan Atique, Aishwarya Ganesan, Ramnatthan Alagappan, **SkyLog: Reducing Cost and Latency in a Multi-Cloud Shared Log** (In submission to NSDI)
- **Xuhao Luo**, Shreesha G. Bhat*, Jiyu Hu*(*equal contribution), Aishwarya Ganesan, Ramnatthan Alagappan, **LazyLog: A New Shared Log Abstraction for Low-Latency Applications** (***SOSP 2024***) ***Best Paper Award, Invited to ACM ToCS***
- **Xuhao Luo**, Ramnatthan Alagappan, Aishwarya Ganesan, **SplitFT: Fault Tolerance for Disaggregated Datacenters via Remote Memory Logging** (***EuroSys 2024***)
- **Xuhao Luo**, Weihai Shen, Shuai Mu, Tianyin Xu, **DepFast: Orchestrating Code of Quorum Systems** (***USENIX ATC 2022***)
- Shreesha G. Bhat, Tony Hong, **Xuhao Luo**, Jiyu Hu, Aishwarya Ganesan, Ramnatthan Alagappan, **Low End-to-End Latency atop a Speculative Shared Log with Fix-Ante Ordering** (***OSDI 2025***)
- Zhiyuan Guo*, Yizhou Shan*(*co-first author), **Xuhao Luo**, Yutong Huang, Yiying Zhang, **Clio: A Hardware-Software Co-Designed Disaggregated Memory System** (***ASPLOS 2022***)

Experience

University of Illinois Urbana-Champaign May. 2021 - Now
Research Assistant *Urbana, IL, USA*

- *LazyLog*: Built a shared log system that offers low-latency appends for latency-critical applications by lazily ordering log entries across multiple storage shards. Used RDMA to achieve μs-level append latency and low end-to-end latency.
- *SplitFT*: Built a new fault-tolerant approach for storage-centric cloud applications by replicating WALs on remote nodes using RDMA. Achieved near-zero performance loss compared with a local file system.
- *SkyLog*: Built a shared log abstraction for multicloud applications, with seamless reconfiguration and reduced egress cost.
- *DepFast*: Built a framework using C++ coroutine to implement and reason about fail-slow tolerant distributed systems in an easy and effective way.

Microsoft Research Jun. 2020 - Sep. 2020
Research Intern *Beijing, China*

- Designed and implemented a task scheduling and dispatching system for distributed ML, and a CUDA-based high-performance inter-GPU communication channel for distributed ML.
- Multi-GPU collective operation(AllReduce, AllGather, Broadcast) throughput outperforms NCCL by at most 18.4%.

Google May. 2026 - Aug. 2026
Software Engineer Intern, Host: Xiaozhou (Steve) Li *Sunnyvale, CA, USA*

- Designed and implemented a general framework to extend Bigtable compaction with user-defined functions. Deployed in production and demonstrated its feasibility to replace certain data processing pipeline.

Amazon Web Service May. 2024 - Aug. 2024
Applied Scientist Intern, Mentor: Prof. George Amvrosiadis, Visiting Scholar *Seattle, WA, USA*

- Investigated and improved ShardStore (storage system used by AWS S3) reclamation, and built a tool to evaluate different reclamation policy.

Amazon Web Service May. 2022 - Aug. 2022
Applied Scientist Intern, Mentor: Shen Li, Principle Engineer *Seattle, WA, USA*

- Improved the performance and reliability of the volume metadata updating workflow for AWS S3 volume metadata cache service with Amazon Quantum Ledger Database (QLDB).

University of California San Diego Sep. 2019 - Dec. 2020
Research Assistant, advised by Prof. Yiying Zhang *La Jolla, CA, USA*

· Implemented a hardware/software co-designed disaggregated memory with FPGA. Worked on the network stack design and implementation.

University of Science and Technology of China Jan. 2019 - May. 2019
Graduation Project, Advisor: Prof. Xi Jin

· Designed and implemented a general 8-bit quantized convolution module on Xilinx Virtex FPGA.

· Developed the TensorFlow C++API for the hardware accelerator using OpenCL. On ResNet-50 CNN, achieved a speedup of 5.17x and a memory usage reduction of 66% compared with the CPU TensorFlow implementation on Xeon E5 2686.

Skills

Language	C/C++, Python, Go, Java, Rust, Haskell, Verilog
Tools/Framework	RDMA, vLLM, SGLang, TensorFlow, Docker, Zookeeper, LLVM

Honors and Awards

- SOSP'24 Best Paper Award Nov 2024
- SOSP'24 Student Travel Grant Nov 2024
- EuroSys'24 Student Travel Grant Apr 2024
- ASPLOS'22 Student Travel Grant Feb 2022
- USTC Class of 2019 Outstanding Graduates May 2019

Services

- EuroSys'26 Shadow PC
- OSDI'23 Artifact Evaluation Committee
- USENIX ATC'23 Artifact Evaluation Committee
- ACM TAAS (Transactions on Autonomous and Adaptive Systems) Invited Reviewer